INCH-POUND
MS21434D
8 November 2022
SUPERSEDING
MS21434C
5 October 2011

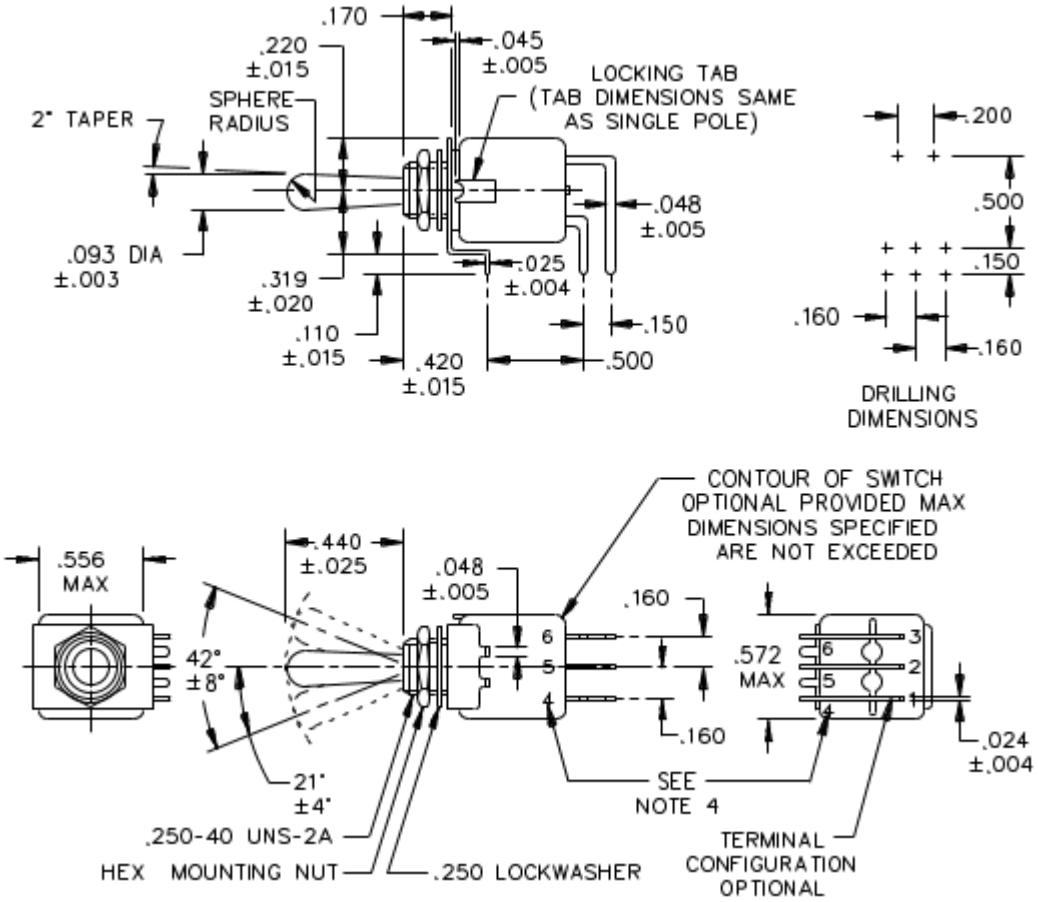
DETAIL SPECIFICATION SHEET

SWITCHES, TOGGLE, POSITIVE BREAK, MINIATURE, TOGGLE SEALED, DOUBLE POLE, PRINTED CIRCUIT BOARD TERMINALS, RIGHT ANGLE MOUNT

Inactive for new design after 5 October 2011

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-DTL-8834.



NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.010 on decimals and ±5° on angles.
3. Terminals need not be marked.
4. Terminal identification is shown for reference purposes only.

FIGURE 1. Dimensions and configuration.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

FSC 5930

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:39Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

For design feature purposes, this specification sheet takes precedence over acquisition documents referenced herein.

115 V ac 60 hertz electrical endurance tests are to be performed at room temperature and pressure.

Switches shall be rigidly mounted to the shock test equipment by means of their terminals and subjected to shock tests of MIL-STD-202-207. The switch shall be electrically and mechanically operative at the conclusion of the test and there shall be no mechanical transfer during the test.

Terminals: Gold plated.

Weight: .0234 pound maximum (8.75 grams).

Terminal strength: Terminals shall be subjected to a five-pound pull for 5 to 10 seconds with the long dimension of the terminal contact vertical. The force shall be applied to the terminal at a point .06 inch (1.5 mm) below the bottom of the switch base, and in a direction parallel to the axis of the toggle lever in its center position. Two outboard terminals shall be tested. However, a terminal shall be tested only once in one direction.

Altitude: 50,000 feet.

Strength of actuator. Lever pivot and lever stop 6 pounds.

Mounting bracket: Brass, tin plated, or coated.

For hardware detail specifications, see appendix of MIL-DTL-8834.

MS dash number -321, -331, -341, and -351, dielectric withstanding voltage: 1,200 V rms at sea level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:39Z
Check the source to verify that this is the current version before use.

TABLE I. Detail requirements.

MS dash no.	Circuit made between terminals as indicated with the toggle lever in these positions: 2/			Current capacity in amperes						Life, low current level switching 5 mV
				Resistive load			Inductive load			
				28 volts dc	115 volts		28 volts dc	115 volts		
	Opposite locking tab side 5/	Center position	Locking tab side 5/		60 Hertz ac	400 Hertz ac		60 Hertz ac	400 Hertz ac	
-211	on 2-3, 5-6	off	on 1-2, 4-5	5	2	3	1	1	2	25 µA 4/
-221		none	off							
-231			on 1-2, 4-5							
-241		off	none							
-271	mom-on 2-3, 5-6		mom-on 1-2, 4-5							
-281	none									
-311	on 2-3, 5-6						3/			
-331 1/	on 2-3, 5-6	on 2-3, 4-5	on 1-2, 4-5							
-341 1/	mom-on 2-3, 5-6	on 2-3, 4-5	mom-on 1-2, 4-5							
-321 1/	none	on 2-3. 5-6	mom-on 1-2, 4-5							
-351 1/	on 2-3, 5-6	on 2-3, 4-5	mom-on 1-2, 4-5							

1/ Delayed action of the switch toggle lever may cause circuit to close or open before snap action mechanism trips.
2/ Direction of movement of internal mechanism is opposite to the direction of the toggle movement.
3/ With time constant of .020 ±.002 second.
4/ Contact resistance not to exceed 50Ω during life, low current level switching.
5/ Non-functional terminals shall not be supplied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:39Z
Check the source to verify that this is the current version before use.

Referenced documents:
MIL-DTL-8834
MIL-STD-202-207

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5930-2022-081)

Review activities:
Army - AR, AT, AV, MI
Navy - EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:39Z
Check the source to verify that this is the current version before use.